

Name: Key

CHE 321; Spring - 2020; Quiz - 4

100 Points

A reversible reaction, $A + B \leftrightarrow C + D$ ($\Delta H_{rxn} = -100$ KJ/mol), is taking place in a CSTR at adiabatic conditions. The rate law for the reaction is:

$$-r_A = k \left(C_A C_B - \frac{1}{K} C_C C_D \right)$$

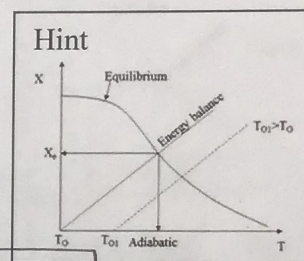
$$K_0 = 3.5673; H_1/R = 600.32 \text{ K}$$

$$K = K_0 \exp\left(\frac{H_1}{RT}\right)$$

1. If for a particular run in the CSTR, the conversion is found to be 0.8, what can you say about the maximum exit temperature possible for the CSTR? [50] [35]

+5 At equilibrium, $r_A = 0$

$$+15 \Rightarrow K = \frac{C_{Ce} \cdot C_{De}}{C_{Ae} \cdot C_{Be}} = \frac{C_{A0}^2 \cdot X_e^2}{C_{A0}^2 (1 - X_e)^2} = \frac{0.8^2}{0.2^2} = 16$$



$$+15 K = K_0 \cdot \exp\left(\frac{H_1}{RT}\right) \Rightarrow 16 = 3.5673 \cdot \exp\left(\frac{600.32}{T}\right) \Rightarrow T_{max} = 400 \text{ K}$$

2. For a CSTR with heat exchanger (typical for the reaction above) in which the above reaction is being carried out, the following equation represents the energy balance. [25]

$$\underbrace{\dot{Q}}_1 + \underbrace{[-\Delta H_{Rx}^o(T_R) - \Delta \hat{C}_p(T - T_R)]F_{Ao}X}_{2} + \underbrace{(-F_{Ao}) \sum_{i=1}^n \Theta_i \tilde{C}_{pi}(T - T_{io})}_{3} = 0$$

This equation has been divided into three parts. Which of these parts represent heat added to the system and which of these represent heat removed from the system? [25]

Heat Added (positive quantities) = 2 +8 each

Heat Removed (negative quantities) = 1, 3

3. For the reaction in question #1, the reactants enter the CSTR at 275 K and the CSTR is run adiabatically. If $F_{Ao} = F_{Bo} = 8$ mol/s, $C_{PA} = C_{PB} = C_{PC} = C_{PD} = 400$ J/(mol $\cdot$ °C), and final temperature is 375 K, find the value of the heat change due to convection/flow. [25]

$$\Delta \text{Heat} = F_{Ao} \Delta T (C_{PA} + C_{PB}) = 8 \frac{\text{mol}}{\text{s}} \cdot 100 \text{ K} \cdot (400 + 400) \frac{\text{J}}{\text{mol} \cdot \text{C}} = 640,000 \text{ J/s} = 640 \text{ kJ/s}$$

4. What is the conversion for problem #3? [15]

$$X = \frac{640 \text{ kJ/s}}{F_{Ao} \cdot -\Delta H_{rxn}} = \frac{640 \text{ kJ/s}}{8 \frac{\text{mol}}{\text{s}} \times 100 \frac{\text{kJ}}{\text{mol}}} = 0.8$$